Young Innovators · Class 9

Rechargeable Power Pack for Robots

Charge it, measure it, regulate it

Lithium cells | Safe charging | Voltage divider | Buck converter



Li-ion cell (protected)



TP4056 charger



Did you know?
The lithium-ion battery won the 2019 Nobel Prize in Chemistry. It powers phones, laptops and electric cars.

Let's think...

Why do robots keep dying mid-race?

Flat again!

Our robots eat AA batteries. They get weaker every minute, the motors slow down, and then we throw the batteries away. Phones charge hundreds of times. Could our robots work like a phone?



Li-ion cell (protected)



Bigger batteries?

Plug into the wall?

Rechargeable pack!

Today: you build a **rechargeable power pack** with a charger, a battery gauge on an OLED, and a regulated output.

Learning objectives

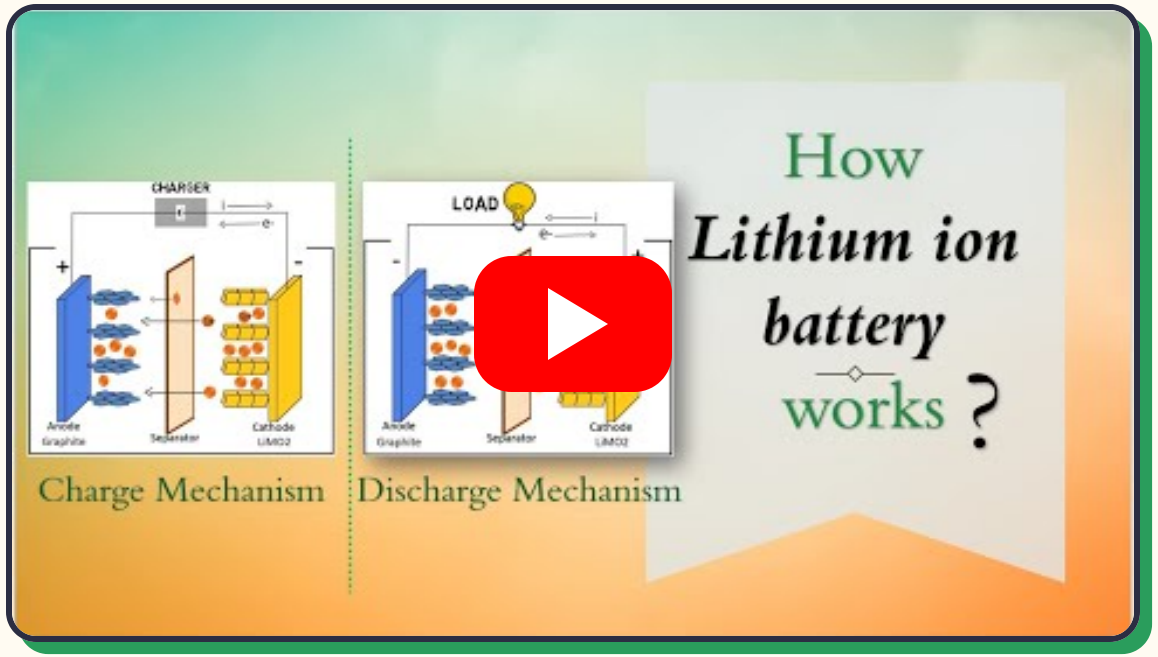
By the end of this class I can…

1	**Battery maths:** explain **voltage**, **capacity (mAh)** and **energy (Wh)** of a lithium cell	**Proof:** I can calculate how long my robot runs on one charge
2	**Charge safely:** charge an 18650 cell safely with a **TP4056** module	**Proof:** the charger LED turns from red to blue/green and I followed every safety rule
3	**Measure:** measure battery voltage with a **voltage divider** and show the % on an OLED	**Proof:** my OLED reading is within 0.1 V of the multimeter
4	**Regulate:** set a **buck converter** to a steady output voltage	**Proof:** the multimeter shows 3.3 V ± 0.05 V at the output

LEARN

Objective 1 · Battery maths · video

Video time!



How a Lithium-Ion Battery Works

Owl WiS · about 3 minutes · youtu.be/YFd0kb9Nwt0

Watch for...

1. Which way do the **lithium ions** move when the battery is charging?
2. What are the **anode** and **cathode** made of?
3. What does the **separator** do, and why is a damaged one dangerous?

Scan to watch

After the video: Explain to your partner, using your hands as the two electrodes, what happens when you charge and then use a battery.

Objective 1 · Battery maths

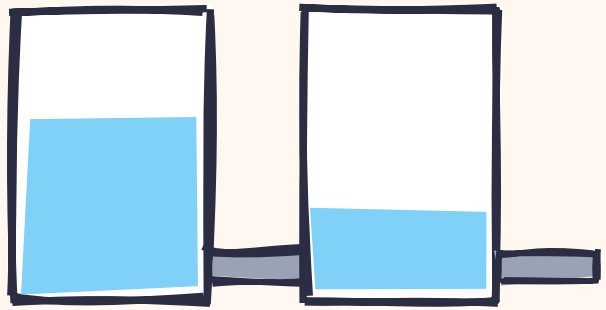
Volts, mAh and watt-hours

The idea

Voltage (V): how hard the battery pushes. One lithium cell: **4.2 V full**, about **3.7 V** in normal use, **3.0 V empty**.
Capacity (mAh): how much charge it holds. 2500 mAh = 250 mA for 10 hours.
Energy (Wh) = V × Ah: 3.7 × 2.5 = **9.25 Wh**.

Run time ≈ capacity ÷ current. Robot draws 500 mA → 2500 ÷ 500 = **5 hours** (less in real life).

Example: a water tank



Voltage is like **how high** the tank is (more pressure). Capacity is **how much water** it holds. A tall thin tank and a short wide tank can hold the same water but push it out differently.

LEARN

Real world

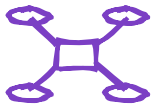
Battery packs everywhere



Phones
One lithium cell, about 4000 to 5000 mAh



Electric cars
Thousands of 18650-style cells joined together



Drones
Light lithium-polymer packs give 20 to 40 min of flight



Solar homes
Store daytime solar energy for the night

? In Pakistan, many homes use UPS batteries during load-shedding. What kind of battery is inside, and how long does it last?

Bigger is not always better
More capacity means more weight. Drone engineers fight for every gram.

Objective 3 · Measure

A voltage divider measures safely

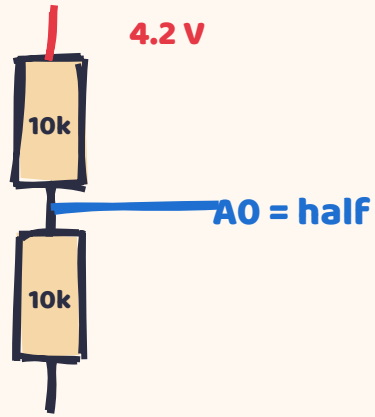
The idea

Two equal resistors in series split a voltage in **half**. Battery up to 4.2 V → the Arduino sees up to **2.1 V** on A0.

In code: **volts = analogRead(A0) × 5.0 ÷ 1023 × 2**
(× 5 ÷ 1023 turns the reading into volts, × 2 undoes the halving.)

Common ground
Battery – must also connect to Arduino GND, or the reading is nonsense.

Example: sharing a pizza fairly



Two equal friends share a pizza: each gets **half**. Two equal resistors share the voltage the same way. If one friend were twice as hungry (twice the resistance), they would get two thirds.

Objective 4 · Regulate

A buck converter steps voltage down

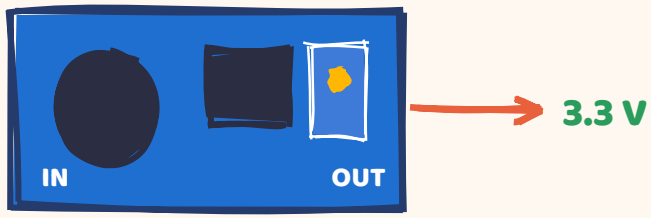
The idea

Different parts want different voltages: ESP32 **3.3 V**, Arduino and servos **5 V**, motors **6 to 12 V**.

A **buck converter** switches on and off very fast to turn a higher voltage into a lower, steady one, wasting little energy (about 80 to 90% efficient). Turn the small screw to set the output.

Only down!
A buck only goes DOWN. One 18650 (3.0 to 4.2 V) cannot make 5 V with it: you would need a BOOST converter or two cells.

Example: a dam with gates



LM2596 buck

A dam holds a high lake, but the gates let out a steady, gentle river. Open them more or less to control the flow. The buck converter is the gate between a high battery voltage and your delicate chip.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Lithium cells store a LOT of energy
2. Battery budget
3. Our power kit (per group)
4. Charge the cell
5. Battery gauge on the OLED
6. Set a steady output
7. Record your results

MAKE

Safety first

Lithium cells store a LOT of energy

Never short-circuit + to – with a wire or metal can make it very hot in seconds.

Never damage don't drop, crush, pierce, or bend a cell. Swollen or dented? Give it to the teacher.

Use the right charger only the TP4056 module, never a direct USB wire. Check polarity before connecting.

Watch it never charge unattended or under books. Touch test: warm is OK, hot means unplug.

Recycle old cells go to battery recycling, not the bin.



Li-ion cell (protected)

Teacher check

Every circuit is checked by the teacher BEFORE the cell goes in the holder.

MAKE

Class game

Battery budget

How to play

1. Each team gets a pretend **2000 mAh** battery (20 paper tokens of 100 mAh).
2. Teacher reads out robot jobs: **motors 1 hour = 600 mAh**, **LEDs 1 hour = 40 mAh**, **Wi-Fi 1 hour = 150 mAh**, **OLED 1 hour = 20 mAh**…
3. Plan a robot's day: pay tokens for each job.
4. Which team can run its robot longest and still do something useful?

Then discuss

What used the most energy? Which jobs could you switch off or slow down?



Li-ion cell (protected)

2000 mAh

Big idea

Engineers design for energy. Motors and radios are usually the hungriest parts.

MAKE

Build it

Our power kit (per group)

18650 cell + holder
The rechargeable energy store

TP4056 module
Safe lithium charger

LM2596 buck
Steps voltage down

Arduino + OLED
Battery gauge display

Multimeter
Checks every voltage

2 × 10k resistors
Voltage divider

Also: breadboard, jumper wires, LED + 220 Ω and buzzer (low battery alarm).

Protected module
Use the TP4056 version with 6 pins (B+, B−, OUT+, OUT−). It also cuts off if the cell gets too empty.

MAKE

Objective 2 · Build step 1 of 3

Charge the cell

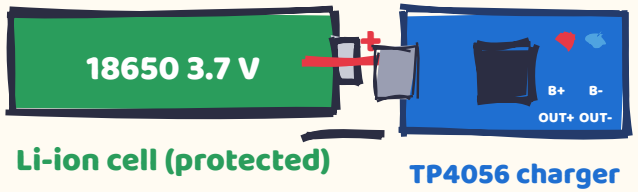
a With the **cell OUT**, check the holder's red wire is +. Multimeter on DC volts.

b Holder + → **B+**, holder − → **B−** on the TP4056.

c Teacher check. Then put the cell in, **flat end to −**.

d Plug a USB cable into the TP4056. Red LED = charging, blue/green = full.

e Measure the cell every 10 minutes and record it in the table on the next test slide.



Check: Voltage rises towards 4.2 V and the LED turns blue/green when full.

Gets hot, smells or hisses? Unplug USB, step back and call the teacher.

MAKE

Objective 3 · Measure · Build step 2 of 3

Battery gauge on the OLED

```
#include <Wire.h>
#include <Adafruit_SSD1306.h>
Adafruit_SSD1306 oled(128, 64, &Wire, -1);

void setup() {
  oled.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  oled.setTextColor(SSD1306_WHITE);
}

void loop() {
  float v = analogRead(A0) * 5.0 / 1023.0 * 2;  // undo divider
  int pct = constrain((v - 3.0) / 1.2 * 100, 0, 100);

  oled.clearDisplay();
  oled.setTextSize(2);
  oled.setCursor(0, 0);  oled.print(v, 2); oled.print(" V");
  oled.setCursor(0, 22); oled.print(pct);  oled.print(" %");
  oled.drawRect(0, 46, 102, 14, SSD1306_WHITE);
  oled.fillRect(1, 47, pct, 12, SSD1306_WHITE);
  oled.display();

  if (v < 3.3) tone(8, 1000, 100);   // low battery beep
  delay(1000);
}
```

Wiring

OLED SDA → A4, SCL → A5, VCC → 5V. Divider middle → A0. Buzzer on pin 8.

Rough %

Lithium voltage does not fall in a straight line, so this % is an **estimate**. Phones use smarter 'fuel gauge' chips.

Check

Compare with the multimeter. Off by more than 0.1 V? Measure your real 5 V pin and use that instead of 5.0.

MAKE

Objective 4 · Build step 3 of 3

Set a steady output

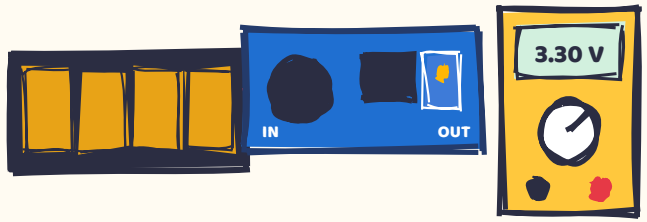
a Connect the 4×AA pack (about 6 V) to the buck **IN+** and **IN–**. Nothing on the output yet!

b Multimeter on the **OUT** pins. Turn the screw slowly.

c Set exactly **3.30 V**. Clockwise usually lowers it on most LM2596 boards.

d Add the LED + 220 Ω as a load. Does the voltage stay steady?

e Measure input and output. Which voltage changed when the batteries were swapped?



Check: Output stays at 3.3 V even when the input changes from 6.2 V to 5.5 V.

Always set the voltage BEFORE connecting a board. 6 V into an ESP32 destroys it.

Test like an engineer

Record your results

Measurement	Value	Unit
Cell at start of charging		V
After 10 / 20 / 30 min		V
Full (LED changed)		V
OLED reading vs multimeter		difference
Buck output, no load / with LED		V

Debug checklist

OLED shows 0 V? Check the divider middle goes to A0 and battery – to GND.
OLED blank? Try address 0x3D.

Charging slows down near 4.2 V. That is normal: the charger switches from constant current to constant voltage.

MAKE

Level up!

Power challenges

Choose your challenge

Run time: measure your robot's current with the multimeter and predict its run time.

Boost: research how an MT3608 boost converter could give 5 V from one cell.

Solar charge: feed the TP4056 from a 6 V solar panel. How long to fill?

Better %: use a lookup table (4.2=100, 3.9=75, 3.8=50, 3.7=25, 3.4=5) instead of a straight line.

Think like an engineer

Why do phones say 1% for a long time, then die suddenly?

Boss level

Build a complete pack: cell + TP4056 + boost + gauge in one box, and power last week's robot.

Objective check

Can I really do it? Prove it.

01	A robot uses 400 mA from a 2000 mAh cell. About how long will it run?	○ ○ ○
02	Name three safety rules for lithium cells.	○ ○ ○
03	Show your OLED gauge and the multimeter. How close are they?	○ ○ ○
04	Show the buck output on the multimeter. Why can't it make 5 V from one 18650?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Power audit: every team states its robot's run time, then tests it next week.

Show what you know

What did we learn?

- [x] Lithium cells: 4.2 V full, 3.7 V normal, 3.0 V empty
- [x] Run time ≈ capacity (mAh) ÷ current (mA)
- [x] Charge only with a proper module, never unattended
- [x] A voltage divider lets the Arduino measure safely
- [x] A buck converter steps voltage down, steadily

At home

Find the mAh rating on a phone or power bank at home. How many times could it charge your robot?



Li-ion cell (protected)



Exit ticket: your gauge shows 3.2 V. Is the cell nearly full or nearly empty, and what should happen?

Next class:
Wireless Doorbell with Visitor Counter!